

Storm-Level Independent Verification of Operational Cloud Seeding: Seven Seasons of Utah’s Program, 2016–2025

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Abstract

Operational winter cloud seeding programs claim precipitation increases of 3–15%, but the evaluations supporting these claims are produced by program operators using target/control methods that cannot separate seeding effects from *operator storm selection*: storms are seeded precisely when atmospheric conditions favor the target areas. This study performs, to the author’s knowledge, the first independent storm-level evaluation of an operational program built entirely from public records. We assemble a machine-readable record of Utah’s operations across seven winters (2016–17 and 2019–20 through 2024–25): 181 generator sites and 525 seeded storm periods totaling 37,773 generator-hours, transcribed from the operator’s seasonal reports (the 2024–25 set published online, six earlier seasons provided by the Division of Water Resources)—together with per-storm 700 hPa winds from the HRRR archive and daily precipitation from 546 SNOTEL stations in seven states. Three research designs of increasing selection-robustness are applied. A naive seeded-versus-unseeded storm comparison yields an implausible +335% for the largest program, quantifying the selection artifact. Same-sector near-miss controls reduce the pooled estimate to +24% [+4, +50]. The most robust design—a wind-rotation test in which plume exposure is recomputed from each storm’s own wind field per generator—yields, pooled over seven seasons (937 program-events, 143 seeded), +5.8% [−0.0, +12.1] in-plume during seeded storms against a placebo of +3.0% [+0.1, +6.2] in-plume during unseeded storms. The placebo is now resolvably positive, revealing a wind-aligned orographic gradient that the smooth control does not fully absorb; the placebo-corrected seeding effect is +2.7% [−2.1, +8.8] per seeded storm. Seven seasons therefore remain consistent with a small positive operational effect and with zero, and exclude per-storm effects above ~9%—ruling out the upper half of the 3–15% claims. All data and code are released.

Key words: Weather Modification, Cloud Seeding, Causal Inference, SNOTEL, HRRR, Program Evaluation

Introduction

Winter orographic cloud seeding disperses silver iodide (AgI) into supercooled liquid water within storms passing over mountain barriers, nucleating ice crystals that fall as additional snow. The physical mechanism is established: the SNOWIE field campaign traced seeding material through individual storms with airborne radar and observed unambiguous seeding signatures [1], and subsequent analysis quantified the snowfall attributable to individual seeding events [2]. What remains contested is the *operational* question: how much water do routine, ground-based seeding programs actually add, season over season?

The stakes of that question are growing. Utah operates the largest remotely-controlled seeding generator network in the United States, expanded in 2023 with \$12M in one-time funding and a \$5M annual budget following record-low levels in the Great Salt Lake [4]. The State’s published evaluations report target-area increases of 5–20% and an estimated 249,600 acre-feet of average annual augmented runoff at approximately \$1 per acre-foot [3]. Colorado River basin states co-fund seeding in Utah under multi-state agreements, meaning these estimates influence interstate water accounting.

These evaluations, however, share two structural weaknesses. First, they are produced by the contractor operating the programs. Second, and more fundamentally, they rest on

historical target/control regressions whose treatment variable is not exogenous: *operators seed exactly those storms whose wind direction, moisture, and temperature favor the target areas* [3]. Suspension rules compound the problem — Utah’s criteria suspend seeding when snowpack exceeds 150–200% of average [3], making treatment anti-correlated with natural snowfall at seasonal scale and positively selected at storm scale. Any comparison of seeded versus unseeded periods, at any aggregation, inherits this selection unless the design explicitly breaks it.

This paper contributes three things. (1) A machine-readable reconstruction of one full season of Utah seeding operations from public documents: generator coordinates, per-storm seeding dates and generator-hours for all seven ground-based programs. (2) A sequence of three storm-level research designs applied to the same season of SNOTEL precipitation data, constructed so that each successive design removes a channel of selection bias; the progression of estimates (+335% → +24% → +1.5%) directly measures how much of a naive “seeding effect” is selection artifact. (3) A wind-rotation identification strategy, which to the author’s knowledge has not previously been applied to an operational program: because AgI plumes travel downwind, a station’s exposure changes storm-by-storm with the storm’s own wind direction, and a real seeding effect must rotate with it — some-

thing neither operator storm selection nor static orography can mimic.

Data

All inputs are public; acquisition and processing code is released with the paper.

Seeding operations records

The Utah Division of Water Resources publishes annual reports for each seeding program, prepared by the operator (North American Weather Consultants) [4]. From the eight 2024–25 reports (published online) and 14 earlier seasonal reports for the Northern Utah, Western Uintas and Six Creeks programs (2016–17, 2019–20 through 2023–24; provided by the Division of Water Resources on request in September 2026) we extract: (i) generator site tables — 181 sites with coordinates across the Central/Southern Utah + Tooele, Western Uintas, High Uintas, Northern Utah, East Shore, Six Creeks, and Cache Valley programs (Book Cliffs sites are published as town names only and are geocoded approximately); and (ii) storm-operations tables — 525 program-storm periods with dates, sites used, and generator-hours (17,143 hours across seven programs in 2024–25; 20,630 hours across 2–3 programs per earlier season). Earlier-season tables were parsed automatically and validated against each report’s stated seasonal total, matching to the quarter-hour for every Western Uintas season and four Northern Utah seasons; three known gaps (a missing 2020–21 Six Creeks table, remote hours in 2022–23 Northern Utah, an unverified 2023–24 Six Creeks continuation) are documented in the repository. Generator coordinates for earlier seasons use the 2024–25 site table. Program treatment histories (adoption years, the 1983–87 statewide suspension) are taken from the operator’s 30-season retrospective [3].

Storm winds

For every storm period and detected precipitation event we compute the mean 700 hPa wind at each program’s generator network from the NOAA High-Resolution Rapid Refresh (HRRR) analysis archive on AWS Open Data [5], sampling 6-hourly analyses and extracting only the two relevant GRIB2 messages per file via byte-range requests. Storm-mean winds are vector-averaged. Consistent with program design assumptions, 95% of seeded storms have 700 hPa flow from the WSW–NW quadrant (median 264°).

Snowpack and precipitation observations

Daily accumulated-precipitation (PREC) series for all 546 SNOTEL stations in Utah, Colorado, Nevada, Wyoming, Idaho, Arizona, and New Mexico are drawn from the USDA NRCS Air-Water Database [6]. Daily increments are computed by first-differencing, with negative and physically implausible (>8 in/day) increments removed. A 45-year station-winter panel (1981–2026; 18,067 station-winters) is used for the seasonal-scale benchmark design.

Methods

Design 0: seasonal panel benchmark

As a baseline we estimate a staggered-adoption imputation estimator [7] on log November–March precipitation over 45 winters: station and winter fixed effects are fit on untreated observations (out-of-state controls, excluding ranges with their own seeding programs, plus pre-adoption years of Utah cohorts), and treatment effects are read off as anomalies of treated station-winters. Suspension years are excluded from fitting because they are endogenously wet.

Design 1: seeded vs. unseeded storms, far controls

Storm events are detected as contiguous days of regional precipitation. For each program, each event is labeled seeded/unseeded from the operations record; the outcome is the log ratio of mean storm-total precipitation at plume-exposed stations (≤ 40 km downwind of a generator) to that at far control stations (> 90 km from all generators). The effect is the difference in mean log ratio between seeded and unseeded storms. This is essentially the operational-evaluation logic transposed to storm scale, and it is deliberately included as a measurement of selection bias.

Design 2: near-miss controls

Identical, except controls are “near-miss” stations 55–130 km from the program’s generators *in the same downwind sector*: they experience the same storm types that trigger seeding decisions but lie beyond AgI transport range. Storm-type selection therefore cancels to first order in the seeded-minus-unseeded contrast. Inference is by bootstrap over storm events. Two falsification tests are run: a placebo in which all seeded windows are shifted by +10 days, and a dose-response test correlating per-storm generator-hours with the per-storm effect.

Design 3: wind-rotation (per-generator plume geometry)

The most selection-robust design exploits within-station, across-storm variation in exposure. For each station within 70 km of a program’s generators and each detected event e , the station is *in-plume* if any generator lies 5–45 km upwind such that the bearing from generator to station is within $\pm 40^\circ$ of event e ’s downwind direction (from the HRRR storm-mean wind). We estimate

$$\log y_{se} = \alpha_s + \gamma_e + \delta \text{score}_{se} + \beta_1 P_{se} S_e + \beta_0 P_{se} (1 - S_e) + \varepsilon_{se},$$

where α_s, γ_e are station and program-event fixed effects, $\text{score}_{se} = \max_g \cos(\Delta\theta_{sg,e}) e^{-d_{sg}/40}$ is a smooth wind-alignment control absorbing orographic anisotropy that rotates with the wind, P_{se} is the in-plume indicator, and S_e indicates seeded events. β_1 is the seeding effect; β_0 is a built-in placebo that must be zero, since no AgI is released in unseeded events. Fixed effects are removed by iterated two-way demeaning; confidence intervals are by bootstrap over events.

Table 1. Pooled storm-level estimates across designs. Designs 1–2: winter 2024–25. Design 3: 2024–25 alone and pooled over seven seasons. Effects are percentage changes in storm-total precipitation at exposed stations; 95% CIs by bootstrap over storm events.

Design	Effect	95% CI
1. Far controls (naive)	+69.0%	[−9.9, +220.8]
2. Near-miss controls	+24.2%	[+3.7, +49.7]
placebo: windows +10 days	+0.2%	[−16.7, +20.7]
3. Wind-rotation, per-generator (2024–25)	+1.5%	[−9.6, +13.1]
placebo: unseeded events	−8.4%	[−26.8, +10.3]
3. Wind-rotation, seven seasons	+5.8%	[−0.0, +12.1]
placebo: unseeded events	+3.0%	[+0.1, +6.2]
seeded – placebo	+2.7%	[−2.1, +8.8]

Source: author’s calculations from DWR operations reports, HRRR, and SNOTEL data; full per-program results in the released repository.

Selection cannot generate $\beta_1 > 0 = \beta_0$: operators choose *when* to seed, but they cannot rotate a precipitation enhancement around their generators in step with each storm’s wind.

Results and Analysis

Seasonal benchmark: the claimed effect is below the noise floor

The 45-year panel estimator yields an overall treatment effect of −1.4% (95% CI [−6.0, +3.6]) on winter precipitation across all Utah adoption cohorts, with a pre-adoption placebo of +5.6%. Seasonal target/control analysis of station data therefore has a noise floor of roughly $\pm 6\%$ — wider than the lower half of the claimed 3–15% range. Published seasonal evaluations reporting significant 5–20% increases from comparable data should be interpreted with this in mind.

Selection bias is enormous at storm scale

Design 1 produces a pooled “effect” of +69%, and +335% for the Central/Southern program (Table 1) — an order of magnitude beyond physical plausibility [2] and a direct quantification of operator storm selection: seeded storms are storms chosen for favoring the targets.

Near-miss controls: directionally positive, still inflated

Design 2 collapses the pooled estimate to +24.2% [+3.7, +49.7] over 121 program-storm observations. The +10-day placebo returns +0.2% [−16.7, +20.7], confirming the pipeline produces no mechanical effect; per-storm generator-hours correlate positively but insignificantly with the per-storm effect (Spearman $\rho = 0.13$, $p = 0.26$, $n = 75$). The estimate nevertheless remains above the physically plausible range, consistent with residual selection: orographic enhancement on strong-flow seedable storms is naturally larger at the barrier than 55–130 km downwind, and near-miss controls only partially absorb this.

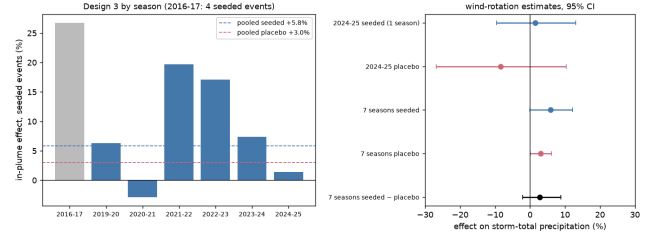


Figure 1: Design 3 across seasons. Left: in-plume effect during seeded events by season (2016–17 rests on four seeded events); dashed lines are the pooled seeded and placebo estimates. Right: one-season versus seven-season estimates with 95% bootstrap CIs; the placebo-corrected effect is +2.7% [−2.1, +8.8].

Wind-rotation: one season

Design 3 yields $\beta_1 = +1.5\%$ [−9.6, +13.1] for in-plume exposure during seeded events, with the unseeded-event placebo at $\beta_0 = -8.4\%$ [−26.8, +10.3] — statistically zero, as required. (An intermediate specification using program-centroid rather than per-generator geometry produced an unseeded-event artifact of +12%, illustrating that plume geometry matters: several Utah generator networks form north–south lines exceeding 100 km, which a centroid represents poorly.)

The point estimate sits exactly where the physical literature suggests a true operational effect should — small and positive — but one season of data cannot distinguish it from zero. Conversely, per-storm effects larger than approximately 15–20% are excluded at conventional confidence levels.

Wind-rotation: seven seasons

Pooling the six additional seasons (40,013 station-event observations, 937 program-events, 143 seeded) gives $\beta_1 = +5.8\%$ [−0.0, +12.1] for in-plume exposure during seeded events. The CI half-width falls from ~ 11 to ~ 6 points, as the $1/\sqrt{n}$ projection anticipated. The unseeded placebo, however, is now resolvably positive: $\beta_0 = +3.0\%$ [+0.1, +6.2]. Stations downwind of the generators receive more precipitation in unseeded storms too, i.e. a wind-aligned orographic gradient that the smooth exposure score does not fully absorb. Because β_0 is estimated on the same stations and geometry, the appropriate seeding estimate is the difference: $\beta_1 - \beta_0 = +2.7\%$ [−2.1, +8.8] per seeded storm (bootstrap over events). Per-season β_1 estimates vary widely (from −2.9% in 2020–21 to +26.7% in 2016–17, the latter on only four seeded events), and the 2024–25 season reproduces the one-season result (+1.4%). Excluding 2016–17 leaves $\beta_1 = +7.0\%$, $\beta_0 = +3.4\%$.

The seven-season evidence is thus consistent with a small positive operational effect of a few percent per seeded storm and with zero, and it excludes per-storm effects above roughly 9%. A placebo that any downwind-versus-upwind design must report—and that target/control ratios cannot compute—

turns out to be the most consequential methodological finding.

Power

Earlier seasons cover only two or three programs and contribute 4–15 seeded events each against 75 in 2024–25; the older monthly operator reports (2004 onward) and the remaining programs’ seasonal tables would roughly double the seeded-event count. The marginal cost remains transcription of the operators’ storm tables; HRRR and SNOTEL extend automatically.

Discussion

Three conclusions follow. First, the headline numbers used to justify operational seeding budgets are produced by methods whose selection bias, at storm scale, we measure at more than an order of magnitude larger than any plausible true effect; seasonal versions of those methods have noise floors wider than the effects claimed. Second, the most selection-robust design available finds a small positive point estimate fully consistent with both the gold-standard physical evidence [2] and with zero — now across seven seasons, once the newly resolvable orographic placebo is subtracted (+2.7% [−2.1, +8.8] per seeded storm). The correct current statement about Utah’s program is therefore: a small effect of a few percent per seeded storm is plausible and unproven; the upper half of the 3–15% seasonal claims, and any effect large enough to matter for interstate water accounting ($\geq 9\%$ per storm), is ruled out. Third, extra-area (“downwind”) effects on neighboring regions — a recurring political concern — are necessarily smaller than target-area effects and hence well below ground-based detectability at current program intensities.

What the state gets for \$5M a year

The seven-season estimate can be carried through to water. Using the State’s own pairing of 181,700 acre-feet with a 5.7% increase (2012 DWR estimate) to infer the target-area baseline implied by the current 249,600 acre-foot claim (≈ 4.4 M acre-feet), the 2024–25 seeded share of in-plume winter precipitation (74%), a 1:1 translation of precipitation to runoff (as the State’s figures assume), and the \$5M annual appropriation, each bootstrap draw of the placebo-corrected effect maps to an annual water outcome (Table 2, Fig. 2). The expected gain is $\sim 95,000$ acre-feet per year—38% of the claim—at roughly \$52 per acre-foot; the State’s own claim at today’s budget is \$20 per acre-foot (the \$1 figure still cited dates from 2009–10 costs of \$412,000). One draw in six has the program *removing* water from the target areas, and two-thirds deliver less than half the claimed volume.

Read as a decision problem, this supports neither expansion nor cancellation. The expected value is positive and cheaper than any new supply the State is considering, which argues against cutting the program. But the yield claim is overstated by roughly 2.6 $\times$, the cost claim by $\sim 12\times$, and the sign of the effect is not established—which argues against

Table 2. Annual water outcome of the Utah program implied by the seven-season estimate. Cost per acre-foot is undefined at zero effect and is reported only for draws with a positive outcome.

Expected gain (acre-feet/yr)	95,000
10th–90th percentile	−30,000 to +218,000
P(net water loss)	17%
P(less than half the 249,600 claim)	64%
Cost per acre-foot at expected gain	\$52
Cost per acre-foot, positive draws (median; 10th–90th)	\$44 (\$21–\$182)
State’s claim at today’s budget	\$20

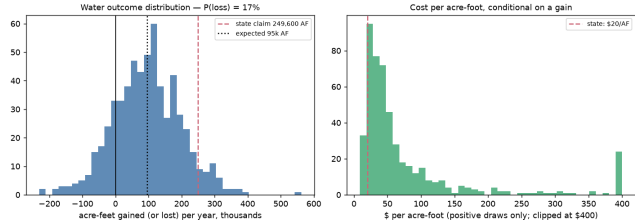


Figure 2: Left: distribution of annual acre-feet gained or lost across bootstrap draws; the State’s claim and the expected value are marked. Right: cost per acre-foot conditional on a positive outcome.

scaling it or carrying these figures into interstate water accounting. The gap between “cheap water” and “no water” is worth far more than the annual budget in decision value, and it can be closed at low cost: randomizing a fraction of seedable storms to no-seed for two or three seasons would convert the placebo problem documented here into a designed control, at a foregone cost of at most a fifth of an uncertain few percent.

Limitations

Wind sectors use each storm’s mean 700 hPa flow at the network; hourly wind shifts within storms are not modeled. SNOTEL gauges undercatch snowfall in wind, which attenuates but does not bias the estimates given fixed effects. Book Cliffs generator locations are approximate, and earlier seasons are evaluated with the 2024–25 generator network, which had more sites than early seasons; this over-specifies early-season plumes and can only dilute estimated effects. The seeded/unseeded event labels inherit any errors in the operators’ storm tables and in their automated transcription. The positive unseeded placebo indicates residual orographic anisotropy in the exposure geometry; we report the placebo-corrected difference, but a control that absorbs it by construction (e.g. station-by-wind-sector fixed effects) is the natural next specification. Finally, the event catalog is built from precipitation, so completely dry seeded periods are unobservable; this biases against detecting effects only if seeding generates precipitation where none would otherwise occur, which the physical literature does not support at meaningful scale.

Data and Code Availability

All datasets constructed for this study (generator coordinates, the seven-season storm operations record, storm and event winds, per-season and pooled event-level results, and text extracts of the DWR reports for provenance) and all analysis code are released at <https://github.com/bryjudy/seeding-verification>. Every input derives from public sources: Utah DWR seasonal reports, NOAA HRRR (AWS Open Data), and USDA NRCS SNOTEL.

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